ARtPUT: Autonomous Rover to Pick Up Trash

A Proof-of-Concept Study of Low-Cost Autonomous Litter Detection and Mechanical Grasping

Bokre Samson

Brendan Amoah-Daniels

Dylan Shattan

Primary Investigator: Dr. Ales Psaker

Governor's School @ Innovation Park

George Mason University

April 26th, 2026

Acknowledgments

We want to thank the people and institutions who made this project possible. Dr. Ales Psaker, our Primary Investigator at the Governor's School @ Innovation Park, mentored us through every stage of this work, from the original design through the hardware troubleshooting and final analysis. His feedback shaped both how we approached the engineering problem and how we built the testing methodology around it.

We also thank the Governor's School @ Innovation Park for the facility access, classroom support, and the chance to do mentorship research as part of our coursework. Finally, we thank George Mason University for hosting the program and providing the educational partnership that made this project possible.

Abstract

Litter in U.S. public spaces generates over 51 billion pieces of debris each year and costs $11.5 billion to clean up (Keep America Beautiful, 2021). Indoor cleaning robots have cut labor costs by up to 80% (Avidbots, 2025), but no equivalent low-cost solution exists for outdoor parks. Systems like the Outdoor Autonomous Trash-Collecting Robot (OATCR) work but require expensive camera hardware (Kulshreshtha et al., 2021). ARtPUT (Autonomous Rover to Pick Up Trash) was built to test a simpler alternative: whether a rover can autonomously detect and pick up litter using low-cost Arduino components and infrared (IR) proximity sensing. The rover uses an Arduino Uno R3, DFRobot Devastator tracked chassis, Sharp IR sensor, and FIT0300 servo-driven gripper. It drives forward while polling the sensor; once readings cross a threshold of 400, the motors halt and the gripper closes. Sixty trials were run across five litter categories at an approximately 50 cm start distance on indoor carpet, with each trial scored Y/N on detection, stopping distance, and grasp. ARtPUT achieved 80% collection success (40/50) across three main categories: crushed bottle (85%), paper ball (80%), and plastic bag (75%), clearing the 70% hypothesis threshold. Detection reached 96%. Two categories failed outright (flat paper disc, 0/5; mandarin peel, 0/5) because their vertical profiles were too low for the IR sensor to detect. Results show low-cost autonomous litter collection is workable, and that object geometry (not detection sophistication) is the main design constraint. Future work will add camera-based vision, a wider gripper, and weatherproofing.

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Introduction and Background

Litter in public spaces is an expensive and persistent environmental problem in the United States. The 2020 National Litter Study found that more than 51 billion individual pieces of debris are generated each year on American roadways, parks, and waterways, and that local governments spend roughly $11.5 billion every year cleaning it up (Keep America Beautiful, 2021). The cost is not only financial. Litter also injures wildlife that ingest or get caught in debris, contaminates soil and water systems, and visibly degrades shared community spaces (Snyder, 2022). Because the problem is widespread, expensive, and ecologically harmful, automating parts of the cleanup process has become an active area of research.

Commercial autonomous cleaning robots have already proven useful in indoor settings. Avidbots (2025), for instance, reports that its industrial floor scrubbers cut labor by up to 80% in warehouse and logistics facilities. The same success has not carried over to outdoor public spaces. Parks, school grounds, and city sidewalks pose much harder problems: uneven terrain, changing lighting, unpredictable obstacles, and litter that varies wildly in material, shape, and weight. As of right now, no widely adopted low-cost autonomous solution exists for outdoor litter collection.

The closest prior work is the Outdoor Autonomous Trash-Collecting Robot (OATCR), described by Kulshreshtha et al. (2021). OATCR shows that vision-based outdoor trash collection works in principle, using a YOLOv4-Tiny convolutional neural network for object recognition combined with an autonomous mobile platform. The system gets strong recognition accuracy on common litter categories under outdoor lighting. The problem is cost. OATCR depends on a high-resolution camera, a processor capable of running deep learning inference,

and a custom-trained model, all of which put it well outside the reach of most schools, volunteer groups, or budget-constrained deployments. The real question is not whether it can be done, but whether it can be done cheaply.

Recent surveys back this up. Memmesheimer et al. (2025) review the broader landscape of cleaning robots in public spaces and propose stakeholder-informed benchmarks, identifying cost, robustness in changing environments, and accessibility as the biggest barriers to wider adoption. The autonomous-systems literature in general has explored similar problems in unmanned aerial vehicle (UAV) navigation, including continuous maneuver control (Li, Ni, & Dressler, 2021) and collision-avoidance strategies for UAV traffic management (Primatesta et al., 2020). These UAV studies are not directly about ground-based trash collection, but they support the broader idea that simple, deterministic control policies, combined with good sensing, can produce reliable autonomous behavior on limited hardware. Other work has shown that simple control logic tends to work reliably when paired with adequate sensing on limited hardware (Boukoberine et al., 2019).

This study targets the cost gap directly. Instead of trying to match the recognition power of vision-based systems, this project asks whether a much simpler sensing approach (infrared proximity detection paired with a basic mechanical gripper) is enough to autonomously detect and collect common kinds of park litter under controlled conditions. The system, named ARtPUT (Autonomous Rover to Pick Up Trash), is built around a DFRobot Devastator tracked chassis (Amazon, 2026), an Arduino Uno R3 microcontroller, and a single Sharp IR analog proximity sensor, a component whose detection reliability in short-range robotics applications has been characterized in prior work (Korba et al., 1994). Two outcomes were expected. First, this study aimed to determine whether the simpler sensing approach could clear a meaningful

collection-rate threshold. Second, this study aimed to identify which litter categories the simpler approach handles reliably and which require more sophisticated sensing or manipulation. This study documents both what worked and what did not, so future low-cost designs have a starting point.

Research Questions

This study investigates two related research questions:

Q1. Can a low-cost Arduino-based rover, using only infrared proximity sensing, autonomously detect and mechanically collect common categories of park litter at a meaningful success rate under controlled indoor conditions?

Q2. Across litter categories that vary in material, shape, weight, and rigidity, which categories are reliably handled by the simpler sensing-and-grasping approach, and which categories define the system's practical limits?

Hypotheses and Computational Model

The alternative hypothesis (H1) states that ARtPUT will successfully detect and collect at least 70% of litter items placed within sensor range under controlled conditions. The null hypothesis (H0) states that the system will fail to reach 70% collection success, meaning the low-cost approach does not produce a useful result. The 70% threshold was picked because it corresponds to the level at which the system would be considered useful as a proof-of-concept rather than a simple demonstration that the system runs.

The system's expected behavior was formalized as a deterministic finite-state machine before testing began. Let $S(t)$ denote the system's state at time t, where $S \in \{DRIVE, STOP, GRASP\}$, and let $I(t)$ denote the IR sensor's analog reading at time t, where $0 \leq I \leq 1023$ (the 10-

bit ADC range of the Arduino Uno). Let $\tau = 400$ denote the empirically tuned detection threshold. The model predicts the following transitions:

- DRIVE → STOP if and only if $I(t) > \tau$
- STOP → GRASP after a fixed 0.5-second servo close delay
- GRASP is terminal (mission complete)

Each trial produces a binary outcome vector ⟨D, P, G⟩ representing detection, positioning, and grasp success. The model also predicts three failure modes: (1) detection failure when an object's reflected IR signal does not exceed τ, (2) grasp failure when an object's width exceeds the gripper's 6 cm jaw span even after detection, and (3) displacement failure when light objects shift before the gripper closes. The data from the 60 trials either back these predictions up or they don't.

Materials and Methods

Hardware Platform

The ARtPUT chassis is a DFRobot Devastator tracked tank platform (Amazon, 2026), chosen for its outdoor-capable rubber tracks, all-metal construction, and standardized mounting holes. The chassis carries two DC gear motors driven through an L298N dual H-bridge motor driver. The motor power supply is an 11.1 V lithium-ion battery pack mounted inside the chassis. The Arduino logic supply is a separate 11.1 V battery, which keeps motor inrush current from causing logic-level voltage drops. The completed assembly is shown in Figure 1. Figure 2 shows the interior chassis layout including the motor driver and battery packs.

The microcontroller is an Arduino Uno R3, picked for its availability, low cost, and how easy it is to program. The original design called for a Raspberry Pi to run camera-based object detection. However, the Raspberry Pi failed during early development and required a redesign of

the sensing pipeline. The Arduino-only system that resulted is simpler, more power-efficient, and more robust against the kind of failures that interrupted the original plan.

Sensing is provided by a single Sharp IR analog proximity sensor mounted on the body of the gripper servo so that the sensor's line of sight tracks the gripper's pointing direction. The actuator is an FIT0300 servo-driven mechanical gripper with a maximum jaw span of about 6 cm. Open and closed positions are commanded as servo angles of 90° and 0°.

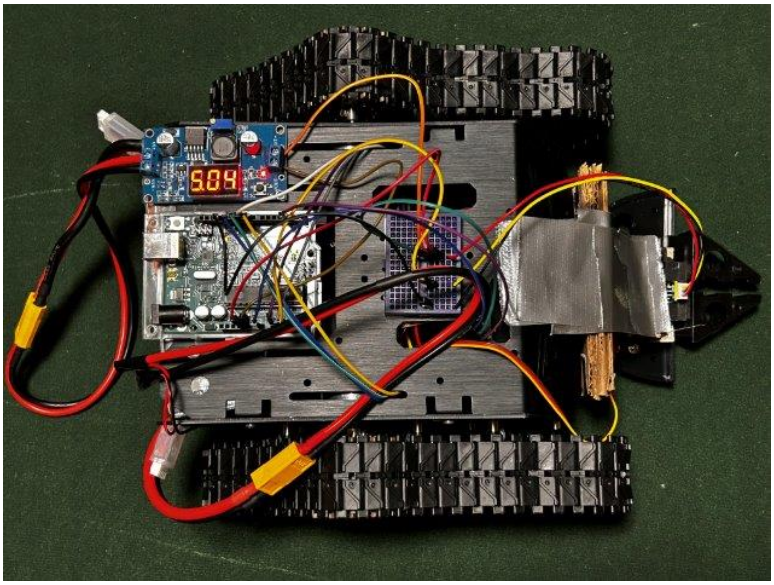

Figure 1. Top view of completed ARtPUT showing the DFRobot Devastator chassis with

mounted Arduino Uno R3, L298N motor driver, and wiring harness.

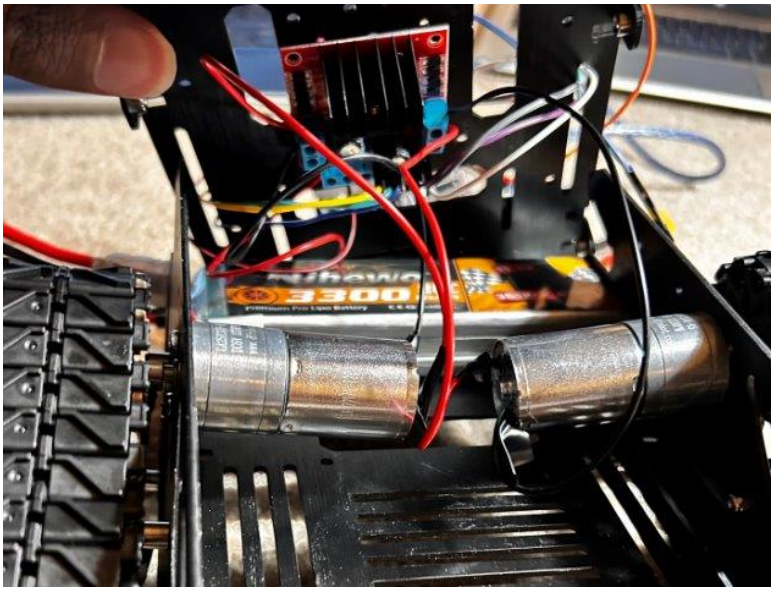

Figure 2. Interior view of the ARtPUT chassis showing the L298N dual H-bridge motor driver, gear motors, and 11.1 V battery packs.

Software and Control Logic

The control software is a single Arduino sketch that implements the state machine described in the Hypotheses section. On power-up, the rover runs a 3-second startup delay so the operator has time to step away. It then enters the DRIVE state and polls the IR sensor on every loop iteration. When the analog reading goes above the threshold value of 400, the rover halts both motors, waits 500 ms for the system to settle, and commands the gripper to close. The full control sequence is illustrated in Figure 3. The threshold value was set empirically during pilot testing. An initial threshold of 350 caused the rover to stop too early, which left the object outside the gripper's reach. The threshold was raised to 400 and locked in for the formal trial set.

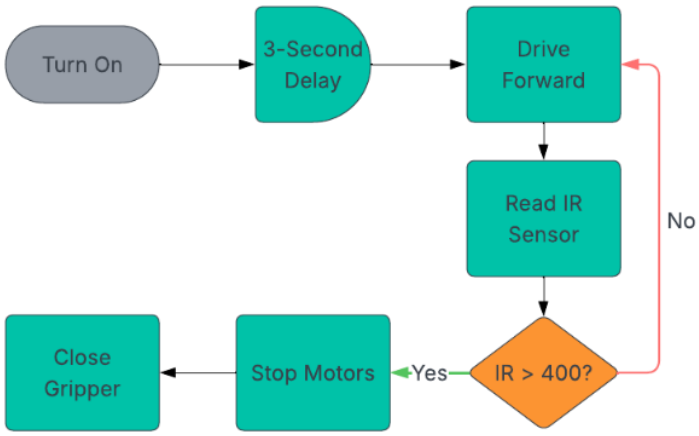

Figure 3. Flowchart of the ARtPUT autonomous detect-stop-grasp control sequence. The rover remains in the DRIVE state until the IR sensor reading exceeds threshold $\tau = 400$, triggering a transition to STOP and then GRASP.

Trial Protocol

All trials were run on a flat, low-pile indoor carpet under consistent overhead lighting. The start position of the rover was marked with electrical tape, and the placement position of the litter item was marked at a fixed distance of approximately 50 cm in front of the rover's nose, also with electrical tape. The trial layout is shown in Figure 4.

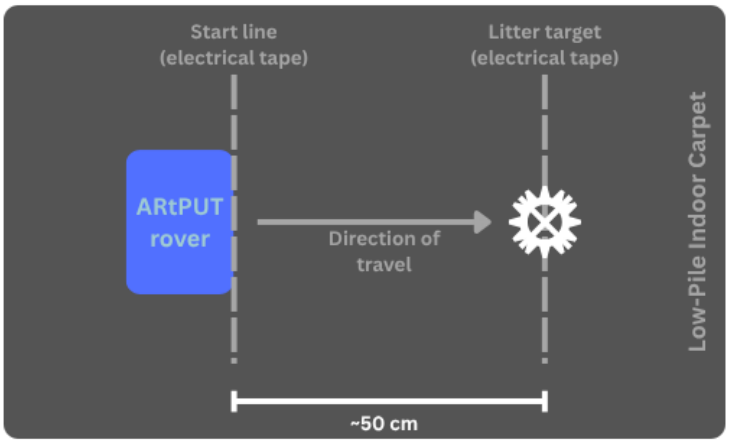

Figure 4. Top-down schematic of the trial setup. The rover was placed at an electrical tape start line on low-pile indoor carpet, approximately 50 cm from a second tape mark indicating the litter placement position. All 60 trials used this configuration.

Before each trial, the rover's orientation was adjusted manually to compensate for a small but consistent leftward drift identified during pilot testing. After placement, the operator hit the rover's power switch and stepped clear. No further input was given during the trial.

Each trial was scored on three binary outcomes. Detection (D) was scored as a yes if the IR sensor triggered a stop while the object was still in the rover's path. Positioning (P) was a yes if the rover halted close enough that the gripper could potentially close around the object. Grasp success (G) was a yes if the gripper physically secured the object. A trial in which all three outcomes were positive was recorded as a full collection success.

Test Items and Sample Sizes

Five categories of litter were tested. Three categories (crumpled paper ball, crushed plastic bottle, and knotted plastic bag) were tested at high replicate counts (10, 20, and 20 trials respectively) and make up the primary outcome set. The other two categories (flattened paper disc and mandarin peel) were tested at five trials each as low-profile limit cases meant to characterize where the simpler sensing approach fails. The reduced sample size for these limit cases was kept because each category showed consistent failure patterns within the first three trials, and additional trials were unlikely to change the categorical finding. Sixty trials were conducted in total.

Data Collection and Analysis

All trials were logged manually by the operator right after each run, including the binary outcomes (D, P, G), any procedural notes (such as object displacement or sensor repositioning between trials), and qualitative observations of failure modes. Per-category success rates were calculated as the count of fully successful trials divided by the trial count. The 70% hypothesis threshold was evaluated both at the overall level (across all 60 trials) and at the primary-category level (across the 50 trials in the main outcome set). Detection rate was computed separately to isolate sensor performance from grasping mechanics. Inferential statistics were not applied, because the binary categorical structure and modest sample sizes did not warrant parametric testing. Instead, the analysis focuses on effect sizes (raw success rates) and pattern-level findings across categories.

Data, Results, and Analysis

Across all 60 trials, ARtPUT successfully detected 50 of the 60 placed objects (83%) and successfully completed the full detection-stop-grasp sequence on 40 of 60 (67%). When restricted to the primary outcome set of three categories (paper ball, crushed bottle, plastic bag; n = 50), the detection rate rose to 96% (48/50) and the full success rate rose to 80% (40/50), exceeding the 70% threshold specified in H1. Per-category outcomes are presented in Table 1 and Figure 5.

Table 1. Results for all 60 trials broken down by litter type. Bottle, paper ball, and plastic bag were the three main categories tested (n = 50). Paper disc and mandarin peel were tested as limit cases to identify the boundaries of what the system can detect and grasp.

Litter Category	Trials	Detected	Grasped	Success Rate

Crushed Bottle	20	20	17	85%
Paper Ball	10	10	8	80%
Plastic Bag	20	18	15	75%
Paper Disc	5	2	0	0%
Mandarin Peel	5	0	0	0%
Total	60	50	40	67% (overall)

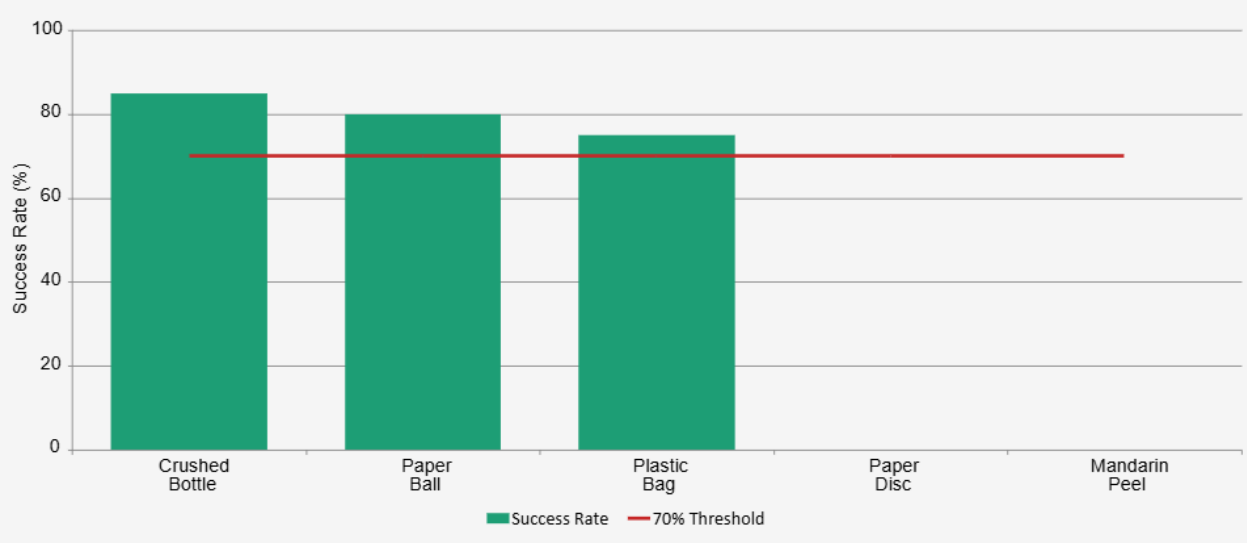

Figure 5. Collection success rate by litter category across all 60 trials. The red line indicates the 70% hypothesis threshold (H1). Primary categories (bottle, paper ball, plastic bag) cleared the threshold; limit-case categories (paper disc, mandarin peel) produced 0% success.

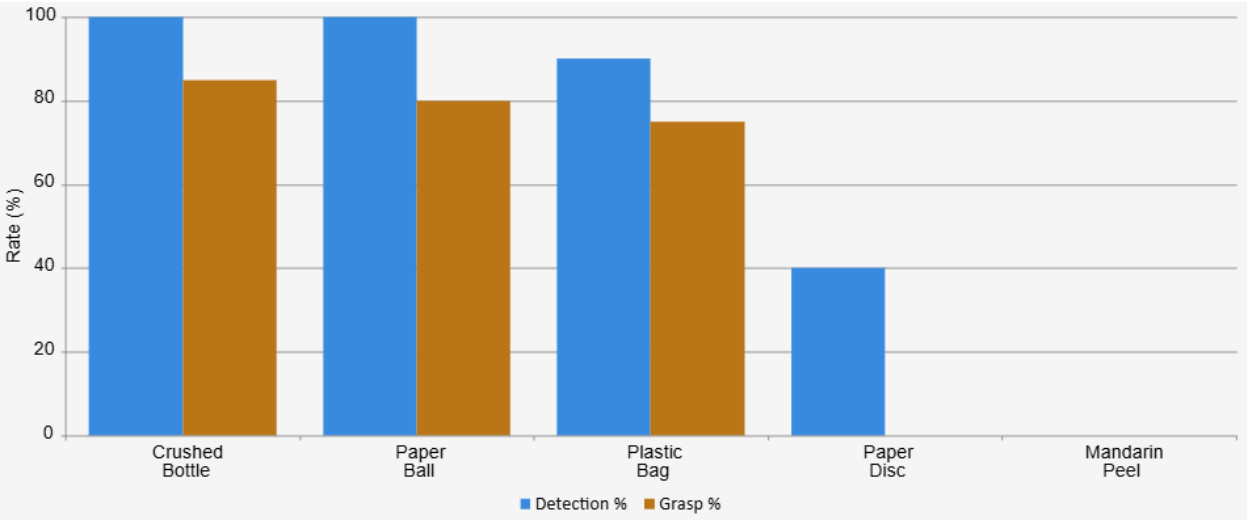

Figure 6. Detection rate versus full collection success rate by litter category. The gap between detection and grasp rates for primary categories isolates mechanical grasping failures from sensing failures.

Performance varied a lot across categories. Crushed plastic bottle, oriented with the cap facing the rover, was the strongest performer at 85% (17/20). Detection on this category was perfect (20/20), and the three observed failures were all mechanical (Figure 7): the gripper jaws hit the rigid cap and the bottle bounced clear of the closing motion. Crumpled paper ball produced an 80% success rate (8/10), with 100% detection and two failures caused by the lightweight paper rolling away from the gripper before the jaws closed completely (Figure 8). Plastic bag produced a 75% success rate (15/20) with 90% detection. Results were uneven early on; only 2 of the first 7 trials succeeded, because the bag was loose and randomly oriented when placed. The bag was knotted tightly and oriented with its narrow end facing the rover, after which all 13 remaining trials succeeded. All 20 trials were retained in the final count rather than removing the early failures, because doing so would have overstated what the system can actually do. The 75% figure is the more honest number, and it also better reflects real-world conditions where litter orientation cannot be controlled.

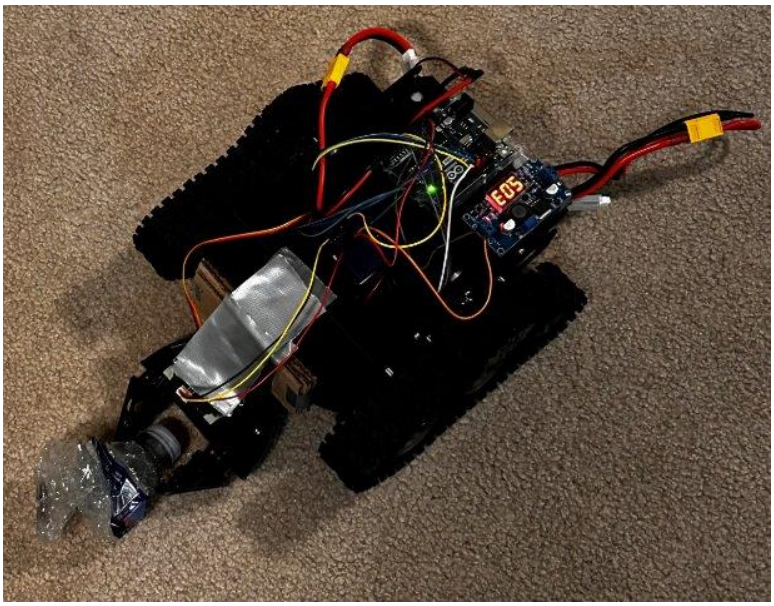

Figure 7. Representative trial showing successful detection and grasp of a crushed plastic water bottle.

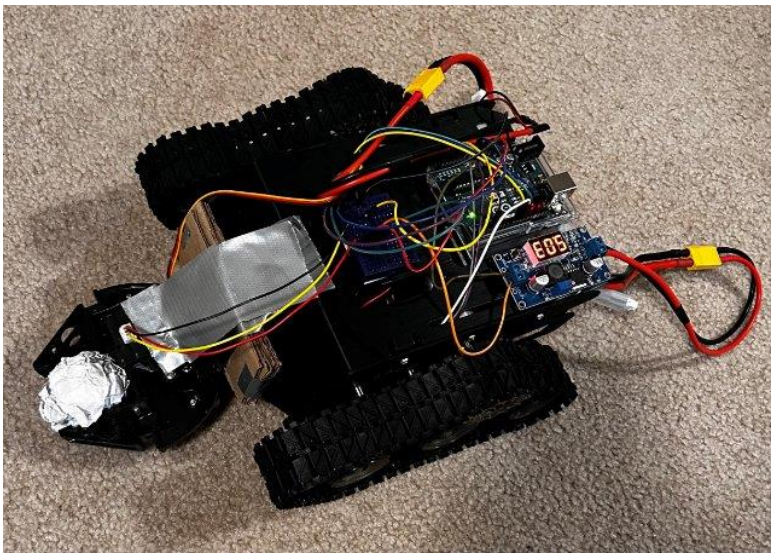

Figure 8. Representative trial showing successful detection and grasp of a crumpled paper ball.

The two limit-case categories produced complete failures. Flattened paper disc was successfully grasped in zero of five trials. Detection was inconsistent (2/5), and even on detected trials, the disc's width exceeded the gripper's jaw span, which kept the jaws from closing around it. Mandarin peel was neither detected nor grasped in any of its five trials. The peel's low vertical

profile (under 1 cm) gave the IR beam no real reflective surface to register against, and the rover drove past the peel in every case. These failures are still informative, as they show exactly where IR-based sensing stops working.

Sample sizes varied across categories and affect the confidence of each reported rate. The paper ball result (8/10, 80%) is the least stable; one more failure would have dropped it to 70%, right at the hypothesis threshold. The bottle and bag categories, each at n = 20, give more reliable estimates. These sample sizes are reasonable for a proof-of-concept study, but future work should increase trial counts, especially for the paper ball category, to get tighter estimates.

Interpretation

The results support H1 across the three primary categories. Across the three main litter categories, ARtPUT cleared the 70% collection threshold by ten percentage points. The 96% detection rate on these categories indicates that the IR sensor reliably picks up objects with enough vertical profile (about 2 cm or greater) within the approximately 50 cm operating range. The gap between detection (96%) and grasp success (80%) indicates that most of the remaining failures were mechanical, not sensor-related. This separation is visible in Figure 6. That matters because a better gripper, not a better sensor, is what the next version actually needs.

The categorical failures of paper disc and mandarin peel, on the other hand, are not fixable through gripper improvements alone. These objects fail the system at the sensing stage. An IR proximity beam fundamentally cannot register objects whose vertical profile falls below the threshold needed for adequate reflection. What this means practically is that object shape matters more than sensor sophistication for systems like ARtPUT. If an object is too flat or too wide, no amount of software tuning fixes it. Camera-based methods like the one used in OATCR

handle flat objects easily because they detect optical contrast rather than physical reflection. Basically, cheaper sensing means less variety of litter can be collected.

The plastic bag results also showed something about methodology. The big improvement in success rate after orientation tuning shows that for flexible, low-mass litter, how the object is presented to the rover matters as much as whether the rover can detect it. Outside the lab, where litter orientations are not controlled, this means flexible bags would need either a more flexible gripper or some way to flip the bag into position before grabbing it. In formal trials, the bag was always set up knotted and narrow-end-forward, and the early failures before that adjustment are included in the count above.

Conclusions and Future Work

This study shows that low-cost autonomous detection and mechanical collection of common park litter is achievable using an Arduino-based rover built around an Arduino Uno R3, an IR proximity sensor, and a servo-driven gripper. The hypothesis that the system would clear a 70% collection rate on primary litter categories was supported by an observed 80% success rate, with 96% sensor detection accuracy. The results also show that object shape (how tall and wide something is compared to the gripper) matters more than how smart the software is.

These results give future builders two things: a cost vs. performance baseline when comparing against more advanced systems like OATCR, and a clear picture of which object types IR sensing can and cannot handle. A few improvements would come directly from what went wrong. First, integrating a low-cost camera module running a lightweight detector like YOLOv4-Tiny would let the system handle flat and low-profile litter while keeping costs low. Second, redesigning the gripper to add a wider jaw span and softer jaw surfaces that can flex around irregular shapes would improve grasping success on irregular and flexible objects. Third,

adding GPS-based path planning would let the rover systematically cover larger outdoor areas, letting the rover actually cover ground instead of just working from a fixed starting point. Finally, weatherproofing the chassis electronics and validating performance on outdoor surfaces (grass, gravel, pavement) needs to happen before this could work anywhere outside a lab.

ARtPUT is not a finished product, but it shows that autonomous litter collection does not have to be expensive. The failures are just as useful as the successes: knowing where IR sensing breaks down is exactly the kind of information a next version would need.

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